which shows that if $y(t)$ is the constant $\bar{y}$ so that $\Delta y(t) = 0$ then the component $F_j(q^{-1})y(t)$ reduces to $\bar{y}$. This, together with the control given by (18), ensures offset-free behaviour by integral action.

3.1. *The control horizon*

The dimension of the matrix involved in (17) is $N \times N$. Although in the non-adaptive case the inversion need be performed once only, in a self-tuning version the computational load of inverting at each sample would be excessive. Moreover, if the wrong value for dead-time is assumed, $\mathbf{G}^T\mathbf{G}$ is singular and hence a finite non-zero value of weighting λ would be required for a realizable control law, which is inconvenient because the "correct" value for λ would not be known *a priori*.

The real power of the GPC approach lies in the assumptions made about future control actions. Instead of allowing them to be "free" as for the above development, GPC borrows an idea from the Dynamic Matrix Control method of Cutler and Ramaker (1980). This is that after an interval $NU < N_2$ *projected control increments are assumed to be zero*,

$$\text{i.e.} \qquad \Delta u(t + j - 1) = 0 \qquad j > NU. \qquad (19)$$

The value NU is called the "control horizon". In cost-function terms this is equivalent to placing effectively infinite weights on control changes after some future time. For example, if $NU = 1$ only one control change (i.e. $\Delta u(t)$) is considered, after which the controls $u(t + j)$ are all taken to be equal to $u(t)$. Suppose for this case that at time t there is a step change in $w(t)$ and that N is large. The choice of $u(t)$ made by GPC is the optimal "mean-level" controller which, if sustained, would place the settled plant output to w with the same dynamics as the open-loop plant. This control law (at least for a simple stable plant) gives actuations which are generally smooth and sluggish. Larger values of NU, on the other hand, provide more active controls.

One useful intuitive interpretation of the use of a control horizon is in the stabilization of nonminimum-phase plant. If the control weighting λ is set to zero the optimal control which minimizes J_1 is a cancellation law which attempts to remove the process dynamics using an inverse plant model in the controller. As is well known, such (minimum-variance) laws are in practice unstable because they involve growing modes in the control signal corresponding to the plant's nonminimum-phase zeros. Constraining these modes by placing infinite costing on future control *increments* stabilizes the resulting closed-loop even if the weighting λ is zero. The use of $NU < N$ moreover significantly reduces the computational burden, for the vector $\tilde{\mathbf{u}}$ is then of dimension NU and the prediction equations reduce to:

$$\hat{\mathbf{y}} = \mathbf{G}_1\tilde{\mathbf{u}} + \mathbf{f}$$

where:

$$\mathbf{G}_1 = \begin{bmatrix} g_0 & 0 & \cdots & 0 \\ g_1 & g_0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & \cdot & & g_0 \\ \vdots & \cdot & & \vdots \\ g_{N-1} & g_{N-2} & \cdots & g_{N-NU} \end{bmatrix} \text{ is } N \times NU.$$

The corresponding control law is given by:

$$\tilde{\mathbf{u}} = [\mathbf{G}_1^T\mathbf{G}_1 + \lambda\mathbf{I}]^{-1}\mathbf{G}_1^T(\mathbf{w} - \mathbf{f}) \qquad (20)$$

and the matrix involved in the inversion is of the much reduced dimension $NU \times NU$. In particular, if $NU = 1$ (as is usefully chosen for a "simple" plant), this reduces to a scalar computation. An example of the computations involved is given in Appendix B.

3.2. *Choice of the output and control horizons*

Simulation exercizes on a variety of plant models, including stable, unstable and nonminimum-phase processes with variable dead-time, have shown how N_1, N_2 and NU should best be selected. These studies also suggest that the method is robust against these choices, giving the user a wide latitude in his design.

3.2.1. N_1: *The minimum output horizon.* If the dead-time k is exactly known there is no point in setting N_1 to be less than k since there would then be superfluous calculations in that the corresponding outputs cannot be affected by the first action $u(t)$. If k is not known or is variable, then N_1 can be set to 1 with no loss of stability and the degree of $B(q^{-1})$ increased to encompass all possible values of k.

3.2.2. N_2: *The maximum output horizon.* If the plant has an initially negative-going nonminimum-phase response, N_2 should be chosen so that the later positive-going output samples are included in the cost: in discrete-time this implies that N_2 exceeds the degree of $B(q^{-1})$ as demonstrated in Appendix B. In practice, however, a rather larger value of N_2 is suggested, corresponding more closely to the rise-time of the plant.

3.2.3. NU: *The control horizon.* This is an important design parameter. For a simple plant (e.g. open-loop stable though with possible dead-time and nonminimum-phasedness) a value of NU of 1 gives generally acceptable control. Increasing NU makes the control and the corresponding output response more active until a stage is reached where any further increase in NU makes little difference. An increased value of NU is more appropriate for complex systems where it is found that good control is achieved when NU is at least equal to the number of unstable or badly-damped poles.

One interpretation of these rules for chosing NU is as follows. Recall that GPC is a receding-horizon LQ control law for which the future control sequence is recalculated at each sample. For a simple, stable plant a control sequence following a step in the set-point is generally well-behaved (for example, it would not change sign). Hence there would not be significant corrections to the control at the next sample even if $NU = 1$. However, it is known from general state-space considerations that a plant of order n needs n different control values for, say, a dead-beat response. With a complex system these values might well change sign frequently so that a short control horizon would not allow for enough degrees of freedom in the derivation of the current action.

Generally it is found that a value of NU of 1 is adequate for typical industrial plant models, whereas if, say, a modal model is to be stabilized NU should be set equal to the number of poles near the stability boundary. If further damping of the control action is then required λ can be increased from zero. Note in particular that, unlike with GMV, GPC can be used with a nonminimum-phase plant even if λ is zero.

The above discussion implies that GPC can be considered in two ways. For a process control *default* setting of $N_1 = 1$, N_2 equal to the plant rise-time and $NU = 1$ can be used to give reasonable performance. For high-performance applications such as the control of coupled oscillators a larger value of NU is desirable.

4. RELATIONSHIP WITH OTHER APPROACHES

GPC depends on the integration of five key ideas: the assumption of a CARIMA rather than a CARMA plant model, the use of long-range prediction over a finite horizon greater than the dead-time of the plant and at least equal to the model order, recursion of the Diophantine equation, the consideration of weighting of control *increments* in the cost-function, and the choice of a control horizon after which projected control increments are taken to be zero. Many of these ideas have arisen in the literature in one form or another but not in the particular way described here, and it is their judicious combination which gives GPC its power. Nevertheless, it is useful to see how previous successful methods can be considered as subsets of the GPC approach so that accepted theoretical (e.g. convergence and stability) and practical results can be extended to this new method.

The concept of using long-range prediction as a potentially robust control tool is due to Richalet *et al.* (1978) in the IDCOM algorithm. This method, though reportedly having some industrial success, is restricted by its assumption of a weighting-sequence model (all-zeros), with an *ad hoc* way of solving the offset problem and with no weighting on control action. Hence it is unsuitable for unstable or nonminimum-phase open-loop plants. The DMC algorithm of Cutler and Ramaker (1980) is based on step-response models but does include the key idea of a control horizon. Hence it is effective for nonminimum-phase plants but not for open-loop unstable processes. Again the offset problem is dealt with heuristically and moreover the use of a step-response model means that the savings in parameterization using a $A(q^{-1})$ polynomial are not available. Clarke and Zhang (1985) compare the IDCOM and DMC designs.

The GMV approach of Clarke and Gawthrop (1975) for a plant with known dead-time k can be seen to be a special case of GPC in which both the minimum and maximum horizons N_1 and N_2 are set to k and only one control signal (the current control $u(t)$ or $\Delta u(t)$) is weighted. This method is known to be robust against overspecification of model-order but it can only stabilize a certain class of nonminimum-phase plant for which the control weighting λ has to be chosen with reasonable care. Moreover, GMV is sensitive to varying dead-time unless λ is large, with correspondingly poor control. GPC shares the robustness properties of GMV without its drawbacks.

By choosing $N_1 = N_2 = d > k$ and $NU = 1$ with $\lambda = 0$, GPC becomes Ydstie's (1984) extended-horizon approach. This has been shown theoretically to be a stabilizing controller for a stable nonminimum-phase plant. Ydstie, however, uses a CARMA model and the method has not been shown to stabilize open-loop unstable processes. Indeed, for a plant with poorly damped poles simulation experience shows that the extended-horizon method is unstable, unlike the GPC design. This is because in this case more than one future output needs to be accounted for in the cost-function for stability (i.e. $N_2 > N_1$).

With $N_1 = 1$, $N_2 = d > k$ and $NU = 1$ with $\lambda = 0$ and a CARIMA model, GPC reduces to the independently-derived EPSAC algorithm (De Keyser and Van Cauwenberghe, 1985). This was shown by the authors to be a particularly useful method with several practical applications; this is verified by the simulations described below where the "default" settings of GPC that were adopted with $N_2 = 10$ are similar to those of EPSAC. Part II of this paper gives a stability result for these settings of the GPC horizons.

Peterka's elegant predictive controller (1984) is nearest in philosophy to GPC: it uses a CARIMA-like model and a similar cost-function, though concentrating on the case where $N_2 \to \infty$. The algorithmic development is then rather different from GPC, relying on matrix factorization and decomposition for the finite-stage case, instead of the rather more direct formulation described here. Though Peterka considers two values of control weighting (λ_1 for the first set of controls and λ_2 for the final δB steps), he does not consider the useful GPC case where $\lambda_2 = \infty$. GPC is essentially a finite-stage approach with a restricted control horizon. This means (though not considered here) that constraints on both future controls and control rates can be taken into account by suitable modifications of the GPC calculations—a generalization which is impossible to achieve in infinite-stage designs. Nevertheless, the Peterka procedure is an important and applicable approach to adaptive control.

5. A SIMULATION STUDY

The objective of this study is to show how an adaptive GPC implementation can cope with a plant which changes in dead-time, in order and in parameters compared with a fixed PID regulator, with a GMV self-tuner and with a pole-placement self-tuner. The PID regulator was chosen to give good control of the initial plant model which for this case was assumed to be known. For simplicity all the adaptive methods used a standard recursive-least-squares parameter estimator with a fixed forgetting-factor of 0.9 and with no noise included in the simulation.

The graphs display results over 400 samples of simulation for each method, showing the control signal in the range -100 to 100, and the set-point $w(t)$ with the output $y(t)$ in the range -10 to 80. The control signal to the simulated process was clipped to lie in $[-100, 100]$, but no constraint was placed on the plant output so that the limitations on $y(t)$ seen in Fig. 4 are graphical with the actual output exceeding the displayed values.

Each simulation was organized as follows. During the first 10 samples the control signal was fixed at 10 and the estimator (initialized with parameters $[1, 0, 0, \ldots]$) was enabled for the adaptive controllers. A sequence of set-point changes between three distinct levels was provided with switching every 20 samples. After every 80 samples the simulated continuous-time plant was changed, following the models given in Table 1. It is seen that these changes in dynamics are large, and though it is difficult to imagine a real plant varying in such a drastic way, the simulations were chosen to illustrate the relative robustness and adaptivity of the methods. The models are given as Laplace transforms and the sampling interval was chosen in all cases to be 1 s.

TABLE 1. TRANSFER-FUNCTIONS OF THE SIMULATED MODELS

Number	Samples	Model
1	1–79	$\dfrac{1}{1 + 10s + 40s^2}$
2	80–159	$\dfrac{e^{-2.7s}}{1 + 10s + 40s^2}$
3	160–239	$\dfrac{e^{-2.7s}}{1 + 10s}$
4	240–319	$\dfrac{1}{1 + 10s}$
5	320–400	$\dfrac{1}{10s(1 + 2.5s)}$

Figure 2 shows the behaviour of a fixed digital PID regulator which was chosen to give reasonable if rather sluggish control of the initial plant. It was implemented in interacting PID form with numerator dynamics of $(1 + 10s + 25s^2)$ and with a gain of 12. For models 1 and 5 this controller gave acceptable results but its performance was poor for the other models with evident excessive gain. Despite integral action (with desaturation) offset is seen with models 3 and 4 due to persistent control saturation.

The first adaptive controller to be considered was the GMV approach of Clarke and Gawthrop (1975, 1979) using design transfer-functions:

$$P(q^{-1}) = (1 - 0.5q^{-1})/0.5 \quad \text{and}$$
$$Q(q^{-1}) = (1 - q^{-1})/(1 - 0.5q^{-1}),$$

with the "detuned model-reference" interpretation. This implicit self-tuner used a k-step-ahead predictor model with $2F(q^{-1})$ and $5\,G(q^{-1})$ parameters and with an adopted fixed value for k of 2. The detuned version of GMV was chosen as the dead-time was known to be varying and the use of $Q(q^{-1})$ makes the design less sensitive to the value of k. Note in particular that for models 2 and 3 the real dead-time is greater than in the two samples assumed.

The simulation results using GMV are shown in Fig. 3; reasonable if not particularly "tight" control was achieved for models 1, 3, 4 and 5. The weighting of control increments (as found in other cases) contributes to the overshoots in the step responses. The behaviour with model 2 was, however, less acceptable with poorly damped transients. Nevertheless, the adaptation mechanism worked well and the responses are certainly better than with the nonadaptive PID controller.

An increasingly popular method in adaptive

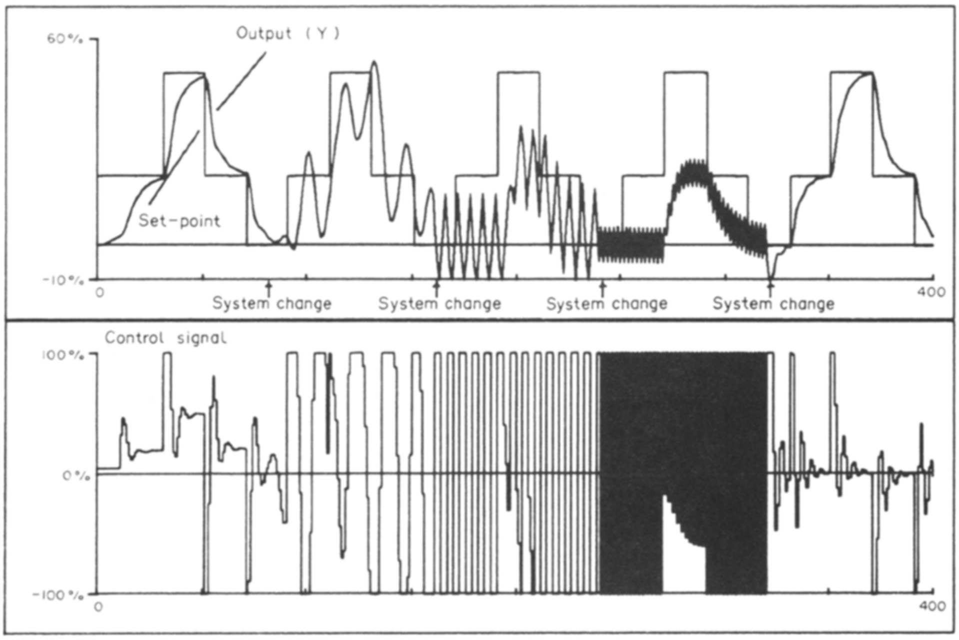

FIG. 2. The fixed PID controller with the variable plant.

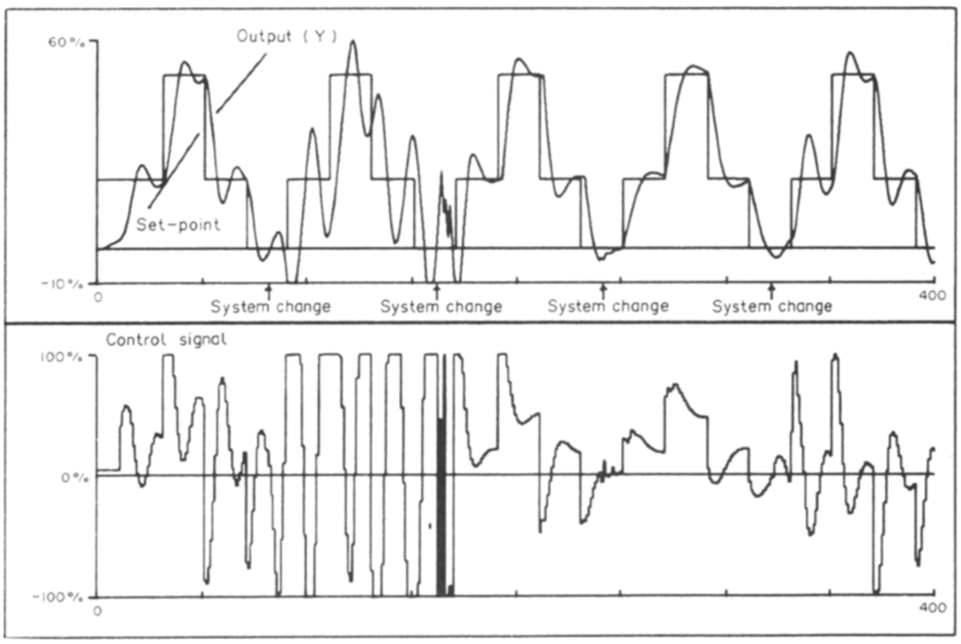

FIG. 3. The behaviour of the adaptive GMV controller.

control is pole-placement (Wellstead *et al.*, 1979) as variations in plant dead-time can be catered for using an augmented $B(q^{-1})$ polynomial. To cover the range of possibilities of dead-time here $2A(q^{-1})$ and $6B(q^{-1})$ parameters were estimated and the desired pole-position was specified by a polynomial $P(q^{-1}) = 1 - 0.5q^{-1}$ (i.e. like the GMV case without detuning). Figure 4 shows how an adaptive pole-placer which solves the corresponding Diophantine equation at each sample coped with the set of simulated models. For cases 1, 2 and 5 the output response is good with less overshoot than the GMV approach. For the first-order models 3 and 4, however, the algorithm is entirely ineffective due to singularities in the Diophantine solution. This verifies the observations that pole-placement is sensitive to model-order changes. In other cases, even though the response is good, the control is rather "jittery".

The GPC controller involved the same numbers of A and B (2 and 6) parameters as in the pole-placement simulation. Default settings of the output and control horizons were chosen with $N_1 = 1$, $N_2 = 10$ and $NU = 1$ throughout, as it has been found that this gives robust performance. Figure 5 shows the excellent behaviour achieved in all cases

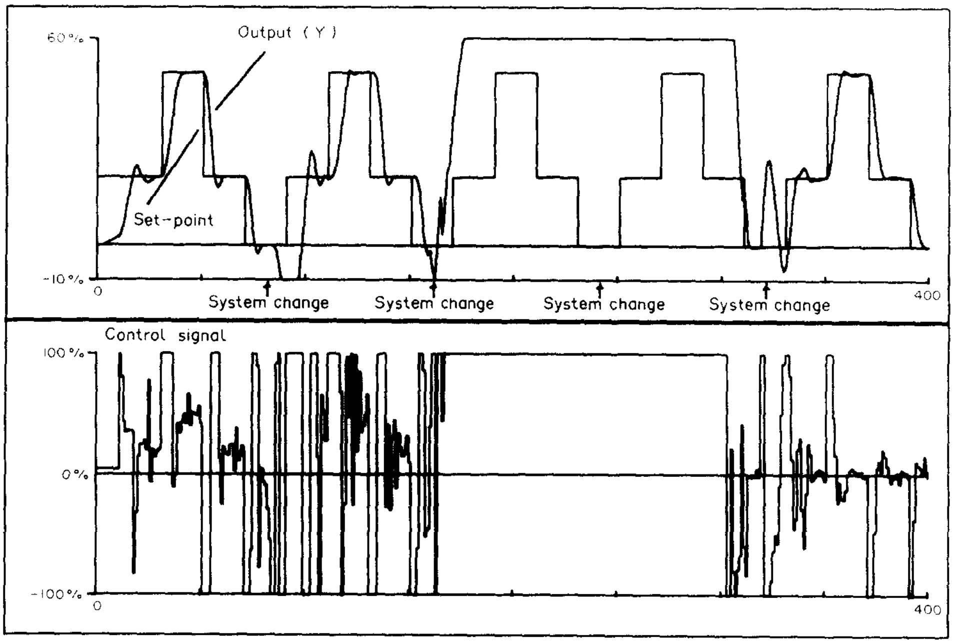

FIG. 4. The behaviour of the adaptive pole-placer.

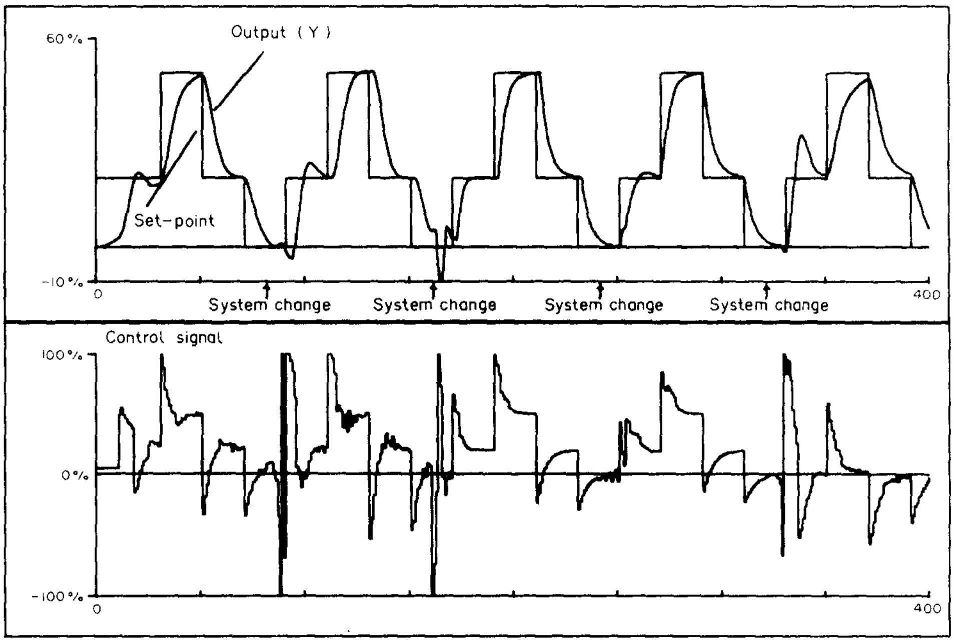

FIG. 5. The behaviour of the adaptive GPC algorithm.

by the GPC algorithm. For each new model only at most two steps in the set-point were required for full adaptation, but more importantly there is no sign of instability, unlike all the other controllers.

6. CONCLUSIONS

This paper has described a new robust algorithm which is suitable for challenging adaptive control applications. The method is simple to derive and to implement in a computer; indeed for short control horizons GPC can be mounted in a micro-computer. A simulation study shows that GPC is superior to currently accepted adaptive controllers when used on a plant which has large dynamic variations.

Montague and Morris (1985) report a comparative study of GPC, LQG, GMV and pole-placement algorithms for control of a heat-exchanger with variable dead-time (12–85 s, sampled at 6 s intervals) and for controlling the biomass of a penicillin fermentation process. The controllers were programmed on an IBM PC in pro-Fortran/proPascal. Their paper concludes that the GPC approach behaved consistently in practice and that "the LQG and GPC algorithms gave the best all-round performance", the GPC method